

Table 4. Cross-Methodological Validation Summary Comparative assessment of hybrid effect detection across statistical frameworks			
Analysis Type	parasiteLoad Results	Ferreira Results	Validation Status ¹
Complete Population	p = 0.017 (Significant)	2 significant effects detected	✓ Partial concordance
Uninfected Subset	p = 0.545 (Not significant)	1 significant effects detected	○ Discordant
Infected Subset	Male-specific: p = 0.038	4 significant effects detected	✓ Strong concordance
¹ ✓ = Methodological concordance, ○ = Methodological discordance or null effects			